

Design Patterns Cheat Sheet - Final Exam Prep

Template Method Pattern

- Defines the skeleton of an algorithm in a method.
- Uses abstract methods for steps that vary.
- Common example: Beverage preparation (Tea, Coffee).
- Supports hooks for optional behavior.
- Promotes code reuse and consistent structure.

Strategy Pattern

- Encapsulates interchangeable behaviors (algorithms).
- Promotes composition over inheritance.
- Allows dynamic change of behavior at runtime.
- Example: Different transaction fee calculations.
- Principle: Program to interface, not implementation.

Adapter Pattern

- Converts one interface into another expected by the client.
- Allows incompatible interfaces to work together.
- Implemented via composition or inheritance.
- Example: Turkey adapted to behave like Duck.

Facade Pattern

- Provides a simplified interface to a complex subsystem.
- Hides complexity but keeps subsystem accessible.
- Improves usability and decouples client from subsystem.
- Example: HomeTheaterFacade to control DVD, projector, etc.

MVC (Model-View-Controller)

- Model: Handles data and business logic (CRUD, rules).

- View: Renders UI, gets data via controller.
- Controller: Interprets input, updates Model/View.
- Promotes separation of concerns.
- Used in web, desktop, and mobile applications.